Cover Letter

Dear Editor,

We submit our manuscript "Structural Determinants of Network Robustness Under Probabilistic Attack: A Cross-Topology Analysis Using Quasi-Monte Carlo Estimation" for consideration in the Journal of Complex Networks.

Most robustness studies assume every attack succeeds with certainty. The RASR framework (Yang & Wang, 2023) addresses this by embedding per-node attack success rates into the robustness definition, yet every published RASR result to date is anchored to a single fixed network. A fundamental question remains unanswered: does the structural class of a network independently determine its robustness under probabilistic attack?

This paper answers that question with the first systematic cross-topology RASR study. Using a complete in-house implementation of the PRQMC algorithm and HBnnsAGP attack strategy across four structurally distinct networks - scale-free, real-world social, small-world, and random - we demonstrate that the fragility ordering Scale-Free < Real-World < Small-World < Random is invariant across every tested ASR level (100% to 50%) and under heterogeneous random ASR. This ordering persists from N=34 to N=10,670, confirming topology class as a stable, independent determinant of probabilistic robustness. We further show that hub-dominated networks exhibit outcome variance roughly twice that of homogeneous networks, introducing a new uncertainty dimension to the RASR framework.

Our contributions include: (1) the first cross-topology RASR analysis, (2) empirical validation of percolation-theoretic fragility predictions under probabilistic attack, (3) a scalability study across four orders of magnitude, and (4) a practical design principle linking degree heterogeneity to hub-protection return on investment.

The manuscript has not been submitted elsewhere and all authors have approved the submission. There are no conflicts of interest to declare. All figures are original and produced by the authors. The study involves no human subjects or animal experiments.

We believe this work is well suited to the Journal of Complex Networks given its focus on structural analysis and mathematical modelling of networks, and we hope it will be of interest to your readership.

We thank you for your time and consideration.

Yours sincerely,

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